

Introduction

The goal of this project is to design, document, and cost a complete cybersecurity infrastructure program for Zenith Orbital Systems. The ZOS Dallas ground station needs to support approximately 200 full-time staff across engineering, mission operations, IT security, and administration. The network has to handle secure wired and wireless access, VoIP communications, server operations, physical security systems, and remote access for personnel at the Austin relay hub or working off-site.

ZOS operations are different from a standard commercial IT environment. We are doing real-time satellite command and control, defense client data relay, and autonomous telemetry processing. The security posture has to match that. Any degradation in ground station capability has a direct impact on active missions and client contracts.

Document Objective

This document covers all the requirements, design decisions, risk findings, policies, plans, and financial details that make up the complete ZOS cybersecurity program. It works as both the board-level funding proposal and the foundational reference document for ZOS security operations going forward. Everything here reflects actual work completed and is ready for implementation once the board approves funding.

Unified Network System Design

The ZOS Dallas network was built from the ground up to meet the specific requirements of a ground station supporting LEO satellite operations. The design balances security with operational continuity so no single point of failure can interrupt mission command capability.

Perimeter defense starts with a Palo Alto PA-5220 Next-Generation Firewall deployed in a high-availability pair. If one unit goes down, the second takes over with zero downtime. The PA-5220 handles the traffic demands of a 200-user organization running real-time data relay. All traffic passes through it with application-layer inspection, intrusion prevention, DNS security, and threat prevention active.

Inside the network, a Cisco Catalyst 9300 Layer 3 switch handles all core routing. Six Cisco Catalyst 9200 Layer 2 access switches distribute connectivity to workstations, offices, and servers. Routing intelligence stays at the core, access-layer connectivity stays at the edge, and that separation makes the network easier to manage and secure.

The network is fully segmented using VLANs. Each VLAN represents a different device type or traffic class and is isolated from everything else unless the firewall explicitly permits it. A compromised workstation cannot reach the server infrastructure, VoIP systems, or physical security devices without going through the firewall first. That is a foundational Zero Trust principle applied at the network layer.

Figure 1: ZOS Dallas Logical Network Topology

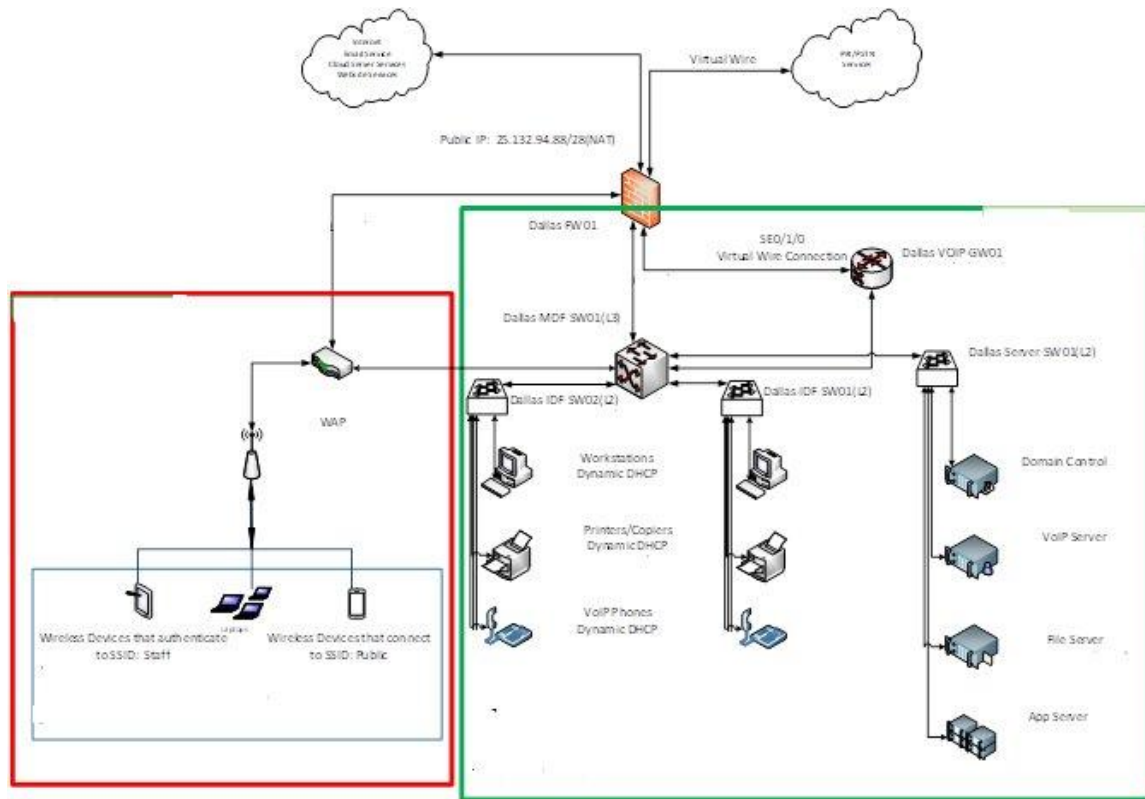


Figure 1: ZOS Dallas Office, Logical Network Topology

VLAN Name	Subnet	VLAN ID	Gateway
Data (Workstations)	10.10.1.0/24	VLAN 10	10.10.1.1
Printers	10.10.2.0/27	VLAN 11	10.10.2.1
WiFi / Wireless	10.10.3.0/24	VLAN 12	10.10.3.1
VoIP Phones	10.10.6.0/24	VLAN 15	10.10.6.1
Servers	10.10.7.0/24	VLAN 20	10.10.7.1
DMZ, Staff	10.110.1.0/30	VLAN 30	10.110.1.1
DMZ, Public	10.110.4.0/30	N/A	10.110.1.5
Firewall Inside	10.10.0.128/29	VLAN 60	10.10.0.130

Network Management	10.10.0.0/25	VLAN 99	10.10.0.1
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Remote access for ZOS personnel working from the Austin relay hub or off-site is handled through Zscaler Zero Trust Exchange. This replaces VPN with a Zero Trust Network Access model where users only get access to the specific applications they are authorized for, not to the broader network. That closes the lateral movement risk that makes traditional VPN a liability.

Physical Topology

The physical network design covers everything deployed across the ZOS Dallas ground station facility. The diagram below shows where all devices are placed, and the component list covers everything in the build.

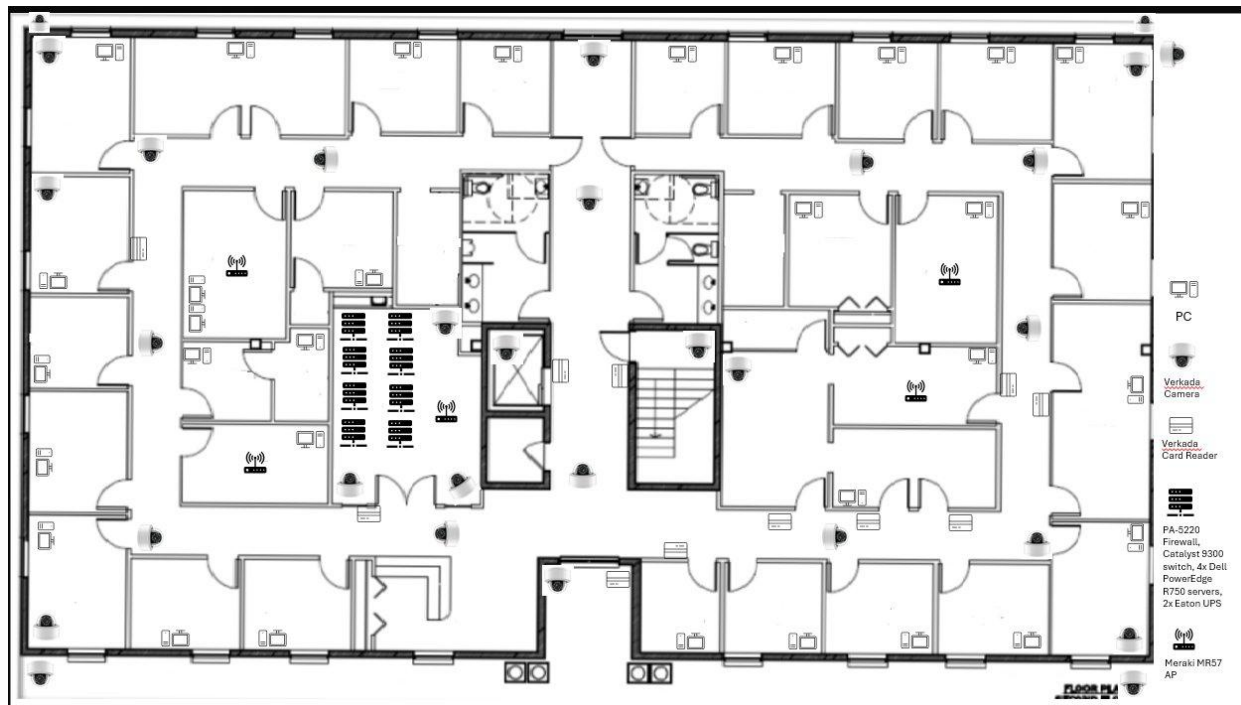


Figure 2: ZOS Dallas Physical Network Topology with Device Placement

- 2x Palo Alto PA-5220 Next-Generation Firewall (HA pair), all licenses
- 2x Cisco Catalyst 9300 48-port Layer 3 Core Switch
- 6x Cisco Catalyst 9200 48-port Layer 2 Access Switch
- 12x Cisco Meraki MR57 Wi-Fi 6E Wireless Access Points
- 4x Dell PowerEdge R750 Servers (16-core, 128GB RAM each)
- 2x Eaton UPS Rack Power Supply units
- 75x Dell Latitude 5550 Laptops (i7, 16GB RAM, 512GB SSD)
- 50x Dell OptiPlex 7010 Desktop Workstations
- 32x Cisco IP Phone 8841 and 8831 Conference Phones
- Verkada dome cameras covering all hallways, entry points, and server rooms
- Verkada card readers at all secured entry points and server room doors
- Cat6a Plenum-rated cabling throughout all work areas and IDF/MDF runs
- 1x biometric safe for media and credential storage

The MDF room sits in the center of the building and houses the core switching stack, the firewall HA pair, all four Dell PowerEdge R750 servers, and the Eaton UPS units. Two IDF distribution points connect to the east and west wings. All cabling runs from workstation locations through conduit to the nearest IDF, which uplinks to the MDF core. All hallways, entry points, and both server rooms are covered by Verkada cameras. Card readers with access logging are on every secured door including the MDF room and both IDF closets. Access logs are reviewed monthly and permissions are audited quarterly.

Wireless Access Point Topology

The ZOS wireless network runs on 12 Cisco Meraki MR57 Wi-Fi 6E access points. These were chosen for their cloud-managed architecture, mesh capability, enterprise security features, and high client density support. Each MR57 supports tri-band operation across 2.4GHz, 5GHz, and 6GHz, which covers every device type in the building.

Each WAP runs two SSIDs. The Staff SSID requires 802.1X authentication validated against the Active Directory domain. Only devices with valid credentials and a current security posture get access. Authenticated staff devices land on VLAN 12 with appropriate access to internal resources. The Public SSID gives internet-only access for visitors and contractors. Public traffic goes straight from the WAP through the firewall to the internet with no connection to the ZOS internal network.

The Meraki cloud management console gives me centralized visibility into every WAP in the building including status, connected clients, traffic volumes, and security events all in one place. All WAP configurations are pushed from the cloud console so nobody can tamper with them locally. Firmware updates deploy automatically during defined maintenance windows.

WAPs are mounted at ceiling level throughout the facility to maximize coverage and minimize obstruction. Placement was validated for both 5GHz and 6GHz coverage across all work areas, conference rooms, and common spaces. There are no dead zones inside the secured facility perimeter.